

Data Visualisation II

Five Design Sheets

Page 1

WHO

- General Australian Public
- Goals: understand where people are online, how free the internet is & where big gaps exist

WHAT

- World Bank: Internet User (% of population) long series (1990-2023; • 2024 backer data)
- Freedom House: Freedom on the net (0-100) + Status (F/PF/NF, 2024)
- Geo: Natural countries (Topo / Geoson)
- Join keys: country ISO3 (country names)
- < 1 MB

GRAPHS IDEAS

- World Map with more Blue if higher % of population on internet
- Another World Map with lighter colour if reached 50% + early, darker if later, orange if not yet
- Bar Chart for status

WHY

- Connectivity $\neq$ Freedom
- Show where access grew, when 50% adaptation was reached and how freedom compares
- Support exploration with clear narrative

Primary Questions

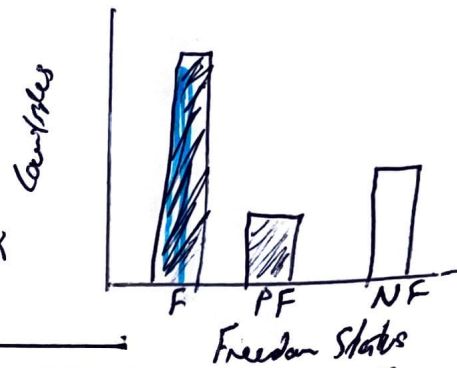
1. Where is access high/low by year?
2. Which countries reached > 50%?
3. How free in 2024?
4. How do access & freedom relate?
5. Who improved most?

FEATURES

- Filter for % online
- Filter for year, latitude, longitude so can zoom
- Tooltip so when hover can see country name, year, % online
- Filter for year reached 50%

PAGES LAYOUT

1. Status bars first - global count of F/PF/NF (sets the stage)
Why First? Gives readers fast sense of freedom mix



2. Map: Global Internet Access (1990-latest)
Controls: Year slider, zoom/lat/lon sliders
Story note beneath

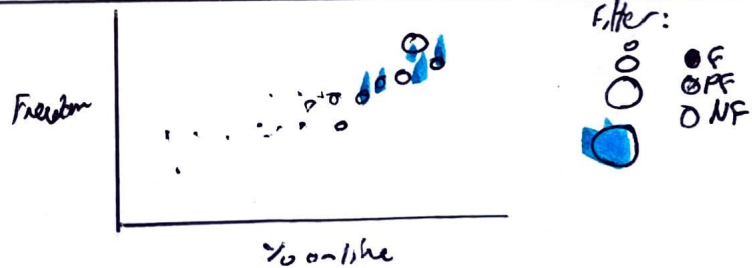


3. Map: Year Country First Reached >50%
Similar controls but no year slider

4. Map: Internet Freedom Score (2024)
• Same projection & colour palette
• Annotation

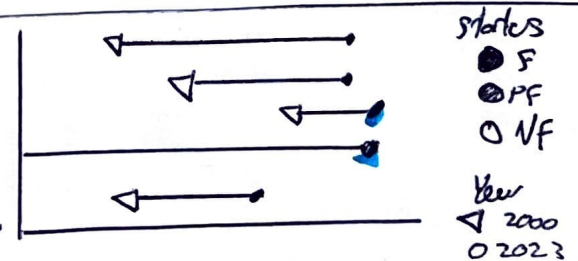
Same map as above (- orange)

5. • Scatter: Access vs Freedom
• X access; Y Freedom
• Annotation
• Bigger dot for higher scores



- 6, 7: Dumbbells: 2000-2023 Access change + Access (2023) vs Freedom (2024)

6. ~~Access~~ Left diamond 2000, right circle 2023
point colour hint by Freedom Status F/PF/NF
7. Dropdown filter by status



8. Top 10 Most Connected: line graph with Top 10 today overlaid
9. Internet Access Overview: Filter select countries & watch by year
10. Access vs Freedom side by side: countries that have both data

PAGE ORDER

1. Global Internet Access - year slider, geographic context
2. Year First Reached > 50% - who crossed threshold & when
3. Status Bars (F/PF/NF) - context for scatter's colour story
4. Internet Freedom Score (2024)
 - align with scatter y-axis distribution
5. Scatter access vs Freedom
 - dotted guides, annotation, lead with relationship
6. Dumbbell: Access 2000 → 2023
 - Growth Narrative
7. Dumbbell: Access vs Freedom
 - easier to see country's gaps
8. Top 10 Lines
 - Leaders converge near 100%
9. Access - Overview
 - user picks country
10. Access vs Freedom Side by Side
 - Comparison made easier

Interaction Plan

- Maps: Year slider, zoom/roll/pan
- Scatter: tooltips, legend explanation
- Dumbbells: tooltips, status filter
- Top-10 lines: hover, legend
- Overview: country dropdown + Brush

COLOUR & LEGENDS

- Single hue like system for consistency
- Freedom score (quantitative) → dark navy to light blue
- Freedom status (F/PE/NE) → 3 tints of same blue family
- Access Quartiles Map → Blue sequential (darker = more online)
- Orange accent for not yet 50%
- Legends: concise, always in top-right, titles

TYPOGRAPHY

- Max line length: 10 words
- Minimal paragraph text per chart
- Chart titles are descriptive

LAYOUT

- Pure CSS
- Fixed Max Width

DATA

- internet access .csv
- Freedom .csv
- world geojson .json
- all json specs readable
- GITHUB

ACCESSABILITY

- Colour-blind friendly (no red + green)
- Canvas renderer

RISKS

- Name mismatches on joins
- Big geo files
- Label overlap

FINAL SEQUENCE

Titles:
Explanation of Dashboard
then

1. Global Internet Access MAP
Year slider + zoom with lat/lon
sets context
then paragraph explanation

2. Year First Reached 50%
Blue by year, orange if not
reaches diffusion & lags
Paragraph explanation

3. Freedom Status Bars
"How Free is the world online"
3 Bars, gives context
Paragraph

4. Internet Freedom (2024) score
single hue blue scale on map
Zoom
Paragraph

5. Access vs Freedom Scatter
Dotted Grids
Annotate
Size by access
Paragraph

6. Connectivity Dumbbell
Growth Arrows
Coloured by status that
spots big movers
Paragraph

7. Country Dumbbell: Access vs Freedom
Percentage lines, filtered by status
Annotate
Paragraph

8. Top 10 Most connected
Leaders of today, historically
Legend on right
Paragraph

9. Internet Access - Overview
Country Dropdown
Brush
Personal Exploration
Paragraph

10. Internet Access vs Freedom
Side-by-side bars
Finishes story
Blue vs Green
Paragraph

What did we learn paragraph

Footer

Finish